Rusiri Ranmini Ilankoon

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PROFESSIONAL SUMMARY

Final-year BSc (Hons) Software Engineering undergraduate at Sri Lanka Technological Campus (SLTC) with experience in software engineering, Artificial Intelligence, machine learning applications, and collaborative software development. Experienced in delivering software projects involving AI, Natural Language Processing, IoT, and MLOps while leading multidisciplinary student teams through effective project coordination, technical documentation, and stakeholder communication. Demonstrated leadership through organizing large-scale university initiatives and managing executive responsibilities within student organizations. Passionate about developing practical software solutions while combining technical knowledge with strong communication, teamwork, and organizational skills.

CORE COMPETENCIES

Software Engineering, Artificial Intelligence, Machine Learning, Software Project Coordination, Technical Documentation, Agile Team Collaboration, Software Requirements Analysis, Public Speaking, Leadership, Cross-functional Collaboration, Event Coordination, Problem Solving

TECHNICAL SKILLS

Programming Languages -
Python, C++, SQL, HTML/CSS

Artificial Intelligence & Data Science -
Machine Learning, Deep Learning, Natural Language Processing (NLP), Explainable AI (SHAP), Data Preprocessing, Feature Engineering, Model Evaluation

Frameworks & Libraries -
TensorFlow, Scikit-learn, Pandas, NumPy, XGBoost, Streamlit

Software Engineering -
Object-Oriented Programming, Software Design, Agile Development, Database Design, REST API Fundamentals, Software Documentation, GitHub

Development & Collaboration Tools -
Visual Studio Code, Google Colab, Jira, Figma

PROJECTS

- Secure Decentralized LoRa Mesh-Based Environmental Sensor Network for Disaster-Prone Areas

  (Final Year Research Project – Ongoing)

  Technologies: LoRa, ESP32, Embedded Systems, IoT, Mesh Networking, Python, C/C++
  - Researching and developing a secure decentralized LoRa mesh communication framework for disaster-prone environments.
  - Designing resilient multi-hop communication between distributed sensor nodes without relying on conventional internet infrastructure.
  - Implementing secure communication mechanisms to improve reliability during emergency situations.
  - Integrating embedded IoT devices for scalable environmental monitoring.

- End-to-End Customer Churn Prediction System with MLOps

  Technologies: Python, Machine Learning, Docker, Git, GitHub, MLOps
  - Developed an end-to-end machine learning pipeline for customer churn prediction following modern MLOps practices.
  - Implemented data preprocessing, model training, evaluation, and workflow management.
  - Applied collaborative software development, version control, and containerization throughout the project lifecycle.
  - Presented the completed system, demonstrating the overall architecture and MLOps workflow.

- AI-Based Patient Readmission Prediction System
  (Ongoing)
  Technologies: Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit
  - Developing an AI-powered healthcare prediction system to identify patients at risk of hospital readmission.
  - Implementing multiple machine learning algorithms including Logistic Regression, Random Forest, Support Vector Machine, and XGBoost.
  - Integrating Explainable AI using SHAP to improve prediction transparency.
  - Developing a Streamlit application for interactive patient risk prediction.

- SMS Spam Detection using Natural Language Processing
  Technologies: Python, TensorFlow, Scikit-learn, Streamlit, Git, GitHub
  - Designed and developed an intelligent SMS spam detection system using Natural Language Processing and Deep Learning.
  - Evaluated multiple machine learning and deep learning models including Naïve Bayes, Logistic Regression, SVM, LSTM, GRU, and CNN.
  - Built a Streamlit application for real-time SMS spam classification.
  - Managed project development using Git and GitHub.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

- Assistant Secretary
  SLTC Media Unit | 2025 – Present
  - Coordinate executive meetings and maintain official organizational documentation.
  - Support strategic planning and administrative operations across multiple committees.
  - Facilitate communication among executive members and contribute to university-wide initiatives.

- Deputy Compering Head
  SLTC Media Unit | 2024 – 2025
  - Hosted official university ceremonies and institutional events.
  - Coordinated stage management and event execution.
  - Strengthened leadership and communication skills through professional event hosting.

- Member
  SLTC Media Unit | 2023 – 2024
  - Supported media coverage, photography, videography, and event coordination.
  - Contributed to organizing Wasath Hiru Udanaya, InnovMind V2.0, IEEE Day, Decode Extreme, Children's Day, Blood Donation Campaign, Sipsari Yathra, Diwali Festival, University Inauguration Ceremonies, Welfare Tailchasers Club Launch, and the Natural Company × SLTC 100 Tree Planting Initiative.

- Assistant Secretary
  IEEE Women in Engineering (WIE) – InspiHer V2.0
  - Assisted in planning, documentation, and communication for the InspiHer V2.0 programme.
  - Supported committee coordination and programme execution.

- Project Secretary
  Withrka 2025 – Inter-School Debating Competition
  - Managed event documentation, committee communication, scheduling, and administrative coordination.
  - Supported the successful execution of the competition organized by the SLTC Media Unit.

EDUCATION

- Bachelor of Science (Honours) in Software Engineering (Reading)
  Sri Lanka Technological Campus (SLTC)
  2023 – Present

- Diploma in English
  IMBS Green Campus (Offered through ABE UK)
  GPA: 3.62

- G.C.E. Advanced Level in Mathematics Stream
  Maliyadeva Model College, Kurunegala

- G.C.E. Ordinary Level
  Maliyadeva Model College, Kurunegala
  6A Passes, 2B Passes, 1C Pass

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

- Assistant Secretary – SLTC Media Unit
- Deputy Compering Head – SLTC Media Unit
- Assistant Secretary – IEEE Women in Engineering (WIE) – InspiHer V2.0
- Event Secretary – Withrka 2025
- University Volunteer
- Public Speaker & Event Host
- Former University Volleyball Player

SOFT SKILLS

Leadership, Project Coordination, Team Collaboration, Communication, Public Speaking, Technical Documentation, Presentation Skills, Event Management, Critical Thinking, Problem Solving, Time Management, Adaptability

REFERENCES

Available upon request